2.1 Measure Preliminaries-1

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Definition 0.1. Let X be a non-empty set. An algebra of sets of X is a non-empty collection $\mathcal{A}$ of subsets of X closed under finite union and complement.

Definition 0.2. A σ-algebra on X is a collection $\mathcal{M}$ of subsets of X satisfying

(i) $\emptyset \in \mathcal{M}$,

(ii) If $E \in \mathcal{M}$, then $E^c = X \setminus E \in \mathcal{M}$,

(iii) If $\{E_n\}_{n\in\mathbb{N}}$ is a sequence of sets in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$.

Now $(X, \mathcal{M})$ is a measurable space and elements of $\mathcal{M}$ are called measurable sets.

1. $X \in \mathcal{M}$
2. If $E_1, \ldots, E_N \in \mathcal{M}$ then $\bigcup_{i=1}^{N} E_i \in \mathcal{M}$
3. If $E_1, \ldots, E_N \in \mathcal{M}$ then $\bigcap_{i=1}^{N} E_i \in \mathcal{M}$
4. If $\{E_i\}_{i\in\mathbb{N}}$ is a sequence of sets in $\mathcal{M}$, then $\bigcap_{i=1}^{\infty} E_i \in \mathcal{M}$
5. If $E, F \in \mathcal{M}$ then $E \setminus F \in \mathcal{M}$

Proof. (a) $\emptyset \in \mathcal{M}$ and $X = (\emptyset)^c \in \mathcal{M}$.

(b) Let $\{E_i\}_{i\in\mathbb{N}}$ be a countable collection of measurable sets where $E_i = \emptyset$ for $i > N$. Hence,

$$\bigcup_{i=1}^{\infty} E_i = E_1 \cup \cdots \cup E_N \in \mathcal{M}.$$

(c) $(\bigcup_{i=1}^{\infty} E_i)^c = \bigcap_{i=1}^{\infty} E_i^c \in \mathcal{M}$

(d) Apply 4 to the collection of sets in 2.

(e) $E \setminus F = E \cap F^c \in \mathcal{M}$ as finite intersection

□

Proposition 0.1. If $\{M_\alpha \mid \alpha \in J\}$ is a collection of σ-algebra on X. Then $\bigcap_{\alpha\in J} M_\alpha$ is a σ-algebra.

Proof. Observe $\emptyset \in M_\alpha$ for all $\alpha \in J$; this implies $\emptyset \in \bigcap_{\alpha\in J} M_\alpha$. If $E \in \bigcap_{\alpha\in J} M_\alpha$, then $E \in M_\alpha$ for all $\alpha \in J$. Hence $E^c \in M_\alpha$ for all $\alpha \in J$, this implies $E^c \in \bigcap_{\alpha\in J} M_\alpha$. If $\{E_n\}_{n\in\mathbb{N}}$ is a sequence of sets in $\bigcap_{\alpha\in J} M_\alpha$, hence $\bigcup_{n\in\mathbb{N}} E_n \in M_\alpha$ for each $\alpha \in J$. Hence $\bigcup_{n\in\mathbb{N}} E_n \in \bigcap_{\alpha\in J} M_\alpha$. □

Definition 0.3. In particular, let $S \subseteq \mathcal{P}(X)$, there exists a unique smallest σ-algebra on X that contains S. This is called σ-algebra generated by S.

Remark 0.1. We show uniqueness by constructing a collection of σ-algebras containing S and then taking an intersection over the family of σ-algebras. By the above proposition, the resulting collection is the smallest σ-algebra containing S.

Definition 0.4. Let X be a topological space. The σ-algebra generated by the topology on X is called **Borel σ-algebra** on X and denoted by $\mathcal{B}_X$. The members of the σ-algebra are called **Borel sets**.

Remark 0.2. Note: F_σ is a countable union of closed sets and G_δ is a countable intersection of open sets. These are examples of Borel sets.

Proposition 0.2. $\mathcal{B}_\mathbb{R}$ is generated by each where $\mathcal{B}_{1\mathbb{R}}$ is a Borel σ-algebra:

(a) open intervals $\mathcal{E}_1$

(b) closed intervals $\mathcal{E}_2$

(c) half-open intervals $\mathcal{E}_3$

(d) open rays $\mathcal{E}_4$

(e) closed rays $\mathcal{E}_5$

Proof. Observe $\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_4$ and $\mathcal{E}_5$ are open or closed sets. For elements $\mathcal{E}_3$, these are G_δ sets, i.e. $(a,b] = \bigcap_{n=1}^\infty (a, b+\frac{1}{n})$. Hence all are Borel sets. If $\mathcal{M}(\mathcal{E}_j)$ is σ-algebra generated by $\mathcal{E}_j$, then $\mathcal{M}(\mathcal{E}_j) \subseteq \mathcal{B}_\mathbb{R}$ for $j = 1, 2, 3, 4, 5$. □

As any open set in $\mathbb{R}$ can be expressed as a union of countable open intervals. Hence $\mathcal{B}_\mathbb{R} \subseteq \mathcal{M}(\mathcal{E}_j)$.

Now observe for any open interval (a, b),

$$\begin{aligned}(a,b) = \bigcap_{n=1}^\infty \left[a+\frac{1}{n}, b-\frac{1}{n}\right] &= \bigcap_{n=1}^\infty \left[a, b+\frac{1}{n}\right] \cap \left[a-\frac{1}{n}, b\right] \\ &= (a,\infty) \cap \bigcap_{n=1}^\infty \left(-\infty, b+\frac{1}{n}\right) \\ &= \bigcap_{n=1}^\infty \left[a-\frac{1}{n}, \infty\right) \cap (-\infty, b)\end{aligned}$$

Hence $\mathcal{M}(\mathcal{E}_j) = \mathcal{B}_\mathbb{R}$ for $j = 1, 2, 3, 4, 5$.

Remark 0.3. Every open set in $\mathbb{R}$ is a countable union of open intervals.

Let U be an open set in $\mathbb{R}$. If $x \in U$ then x is rational or irrational. Let x be rational then construct

$$I_x = \bigcup_{x \in I} I$$

where I are open intervals containing x and contained in U. If x is not a rational for some $\varepsilon > 0$, $x \in (a-\varepsilon, a+\varepsilon) \subseteq U$, choose a rational $q \in (a-\varepsilon, a+\varepsilon)$ and observe

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Let $x \in I_q$. Hence for $x \in U$ there exists $q \in \mathbb{Q}$ such that $x \in I_q$. This implies $U \subseteq \bigcup_{q\in\mathbb{Q}} I_q$. Observe that $I_q \subseteq U$ for all $q \in \mathbb{Q}$ hence

$$\bigcup_{q\in\mathbb{Q}} I_q = U \quad \text{where } I_q \text{ are open intervals.}$$

Definition 0.5. A family of sets $\mathcal{R} \subseteq \mathcal{P}(X)$ is called a *ring* if it is closed under finite union and difference (i.e. if $E_1, \ldots, E_n \in \mathcal{R}$ then $\bigcup_{i=1}^n E_i \in \mathcal{R}$ and if $E, F \in \mathcal{R}$ then $E \setminus F \in \mathcal{R}$). A ring that is closed under countable unions is called a σ-ring.

Proposition 0.3. If $\mathcal{R}$ is a σ-ring, then it is closed under countable intersection.

Proof. For $F \in \mathcal{R}$ observe $\emptyset = F \setminus F \in \mathcal{R}$. Let $\{E_i\}_{i=1}^\infty$ be a collection of sets belonging to $\mathcal{R}$. Then

$$\bigcap_{i=1}^\infty E_i = \bigcap_{i=1}^\infty (F \setminus (F \setminus E_i)) = F \setminus \bigcup_{i=1}^\infty (F \setminus E_i) \in \mathcal{R},$$

since $\mathcal{R}$ is a σ-ring. □

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To $\mathcal{R}$. If any of them are pairwise disjoint $\bigwedge_{i=1}^n E_i = \emptyset \in \mathcal{R}$. Assume $\{E_i\}_{i=1}^n$ are not pairwise disjoint. As $\mathcal{R}$ is a σ-ring, let $\bigcup_{i=1}^n E_i = E \in \mathcal{R}$. Let $E_i' = E \setminus E_i \in \mathcal{R}$ for $i \in \mathbb{N}$. Now $E' = \bigcup_{i=1}^n E_i' \in \mathcal{R}$. Observe $\bigwedge_{i=1}^n E_i = E \wedge E' \in \mathcal{R}$.

□

Definition 0.6. Let X be a set equipped with a σ-algebra $\mathcal{M}$. We define a **countably additive measure** m such that:

(i) $m : \mathcal{M} \to [0, \infty]$ is a well-defined function.

(ii) $m(\emptyset) = 0$.

(iii) If $\{E_j\}_{j=1}^\infty$ is a sequence of disjoint sets, then

$$m\left(\bigcup_{j=1}^\infty E_j\right) = \sum_{j=1}^\infty m(E_j).$$

Observe that (iii) implies finite additivity by taking $E_j = \emptyset$ for $j > n$.

We define $(X, \mathcal{M}, m)$ as a **measure space**.

Definition 0.7. If $m(X) < +\infty$, we call m a **finite measure**.

Definition 0.8. If $X = \bigcup_{j=1}^\infty E_j$ where $E_j \in \mathcal{M}$ and $m(E_j) < +\infty$ for all j, then m is a **σ-finite measure**.

Definition 0.9. More generally, E is a **σ-finite set** if $E = \bigcup_{j=1}^\infty E_j$, $E_j \in \mathcal{M}$ and $m(E_j) < +\infty$ for all j.

Definition 0.10. If for each $E \in \mathcal{M}$ with $m(E) = \infty$ and there exists $F \subseteq E$ such that $0 < m(F) < \infty$, then m is *semifinite*.

Every σ-finite measure is semifinite.

Proof. If $m(E) = \infty$, then observe $E = E \cap X = \bigcup_{j=1}^{\infty}(E \cap E_j)$. This implies $E \cap E_j \subseteq E_j \subseteq X$ for some j. We know $m(E_j) < +\infty$, hence $m(E \cap E_j) < m(E_j) < +\infty$. Hence we have found a subset of E, specifically $E \cap E_j$, which has measure of $E_j \cap E$ to be finite. $\square$

Example 0.1. Let X be any set and $x_0 \in X$ be a fixed point and $\mathcal{M} = \mathcal{P}(X)$. Let $m : \mathcal{M} \to \mathbb{R}$

$$m(E) = \begin{cases} 1 & \text{if } x_0 \in E \\ 0 & \text{if } x_0 \notin E \end{cases}$$

This is called the **Dirac measure**.

Example 0.2. Let X be any set and $\mathcal{M} = \mathcal{P}(X)$. Let $m : \mathcal{M} \to [0, \infty]$

$$m(E) = \begin{cases} |E| & \text{if } |E| < +\infty \\ \infty & \text{otherwise} \end{cases}$$

where $| \cdot |$ is the cardinality function and m is called the **counting measure**.

Example 0.3. Let X be any set and $\mathcal{M} = \mathcal{P}(X)$. Let $m : \mathcal{M} \to [0, \infty]$

$$m(E) = +\infty \quad \text{for all } E \subset X.$$

Lemma 0.1. Suppose X be a set and $\mathcal{M}$ is a σ-algebra equipped on X. Let $m : \mathcal{M} \to [0, \infty]$ be a measure, then

(i) [**Monotonicity**]: If $A, B \in \mathcal{M}$ such that $A \subseteq B$, then

$$m(A) \leq m(B).$$

(ii) If $\{A_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}$, then

$$m\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} m(A_n).$$

Proof. (i) If $B \supseteq A$, then

$$m(B) = m(A \cup (B \setminus A)) = m(A) + m(B \setminus A) \geq m(A).$$

(ii) Let $E_1 = A_1$, and $E_k = A_k \setminus \bigcup_{i=1}^{k-1} A_i$ for $k \geq 1$. Then observe all E_k are disjoint to each other, hence

$$m\left(\bigcup_{k \in \mathbb{N}} E_k\right) = \sum_{k \in \mathbb{N}} m(E_k).$$

Note that

$$\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{k \in \mathbb{N}} E_k \quad \text{and} \quad m(E_k) \leq m(A_k) \quad \text{for all } k \in \mathbb{N}.$$

Hence we get

$$m\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} m(A_n).$$

□

Lemma 0.2. Let $(X, \mathcal{M}, \mu)$ be a measure space.

1. **Continuity from below:** If $\exists\, E_j \uparrow_j^\infty = 1 \in \mathcal{M}$ and $E_1 \subseteq E_2 \subseteq \cdots$, then

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \lim_{j\to\infty} \mu(E_j).$$

2. **Continuity from above:** If $\exists\, E_j \downarrow_j^\infty = 1 \in \mathcal{M}$ and $E_1 \supseteq E_2 \supseteq \cdots$, then

$$\mu\left(\bigcap_{j=1}^\infty E_j\right) = \lim_{j\to\infty} \mu(E_j),$$

and $\mu(E_1) < +\infty$.

Proof. (a) Set $E_0 = \emptyset$. Observe $\bigcup_{j=1}^\infty E_j = \bigcup_{j=1}^\infty E_j \setminus E_{j-1}$. Hence,

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \mu\left(\bigcup_{j=1}^\infty (E_j \setminus E_{j-1})\right)$$

$$= \sum_{j=1}^\infty \mu(E_j \setminus E_{j-1}) = \lim_{n\to\infty} \sum_{j=1}^n \mu(E_j \setminus E_{j-1})$$

$$= \lim_{n\to\infty} \mu(E_n).$$

(b) Let $F_j = E_1 \setminus E_j$. Observe $F_1 \subseteq F_2 \subseteq \cdots$ and $\bigcup_{j=1}^\infty F_j = E_1 \setminus \bigcap_{j=1}^\infty E_j$. Then,

$$\mu(E_1) = \mu\left(\bigcap_{j=1}^\infty E_j\right) + \lim_{n\to\infty} \mu(F_n).$$

□

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$$\mu(E_i) = \mu\left(\bigcap_{i=1}^\infty E_j\right) + \lim_{n\to\infty} \mu(E_i \setminus E_j)$$

$$= \mu\left(\bigcap_{i=1}^\infty E_j\right) + \mu(E_i) - \lim_{n\to\infty} \mu(E_j).$$

This implies

$$\mu\left(\bigcap_{i=1}^{\infty} E_j\right) = \lim_{n\to\infty} \mu(E_j),$$

as $\mu(E_i) < +\infty$, we can subtract it from both sides.

Proof. □ □

Definition 0.11. If $(X, \mathcal{M}, \mu)$ is a measure space, a set $E \in \mathcal{M}$ such that $\mu(E) = 0$ is called a null set.

If a statement is true about point $x \in X \setminus E$ for some null set E, then we say the statement holds almost everywhere (a.e). (More precisely, we have μ-null set and μ-a.e.)

Corollary 0.1. Any countable union of null sets is a null set, and subsets of null sets belonging in $\mathcal{M}$ are null sets.

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Definition 0.12. If a measure whose domain includes all subsets of a null set then the domain is called **complete**.

Theorem 0.1. Suppose that $(X, \mathcal{M}, \mu)$ is a measure space. Let

$$\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$$

and

$$\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subseteq N \text{ for some } N \in \mathcal{N}\}.$$

Then $\overline{\mathcal{M}}$ is a σ-algebra and there is a unique extension $\overline{\mu}$ of μ to complete measure $\overline{\mathcal{M}}$.

Proof. **(a) $\overline{\mathcal{M}}$ is a σ-algebra**

Since $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions, so is $\overline{\mathcal{M}}$. If $E \in \mathcal{M}$ and $F \subseteq N \in \mathcal{N}$ then $E \cup F \in \overline{\mathcal{M}}$. Now assume $E \cap N = \emptyset$ otherwise replace F by $F \setminus E$ and N by $N \setminus E$. Then

$$E \cup F = (E \cup N) \cap (N^c \cup F),$$

so

$$(E \cup F)^c = (E \cup N)^c \cup (N^c \cup F)^c$$

and observe □

$$(EUN)^c \in \mathcal{M} \text{ and } N \setminus F^c = N \setminus F \subseteq N \text{ and } N \setminus F \in \mathcal{N}. \text{ Hence } (EUF)^c \in \overline{\mathcal{M}}.$$

Proof. (cb) Define $\overline{\mu} : \overline{\mathcal{M}} \to [0, \infty]$, such that

$$\overline{\mu}(EUF) = \mu(CE).$$

We want to show well definedness. Let $E_1UF_1 = E_2UF_2$ where $E_1, E_2 \in \mathcal{M}$ and $F_1, F_2 \in \mathcal{N}$. Now,

$$E_1 \subseteq E_1UF_1 = E_2UF_2 \subseteq E_2UN_2.$$

This implies

$$\mu(CE_1) \leq \mu(E_2) + \mu(N_2) = \mu(CE_2).$$

If $E_2 \subseteq E_1 UN$, and this implies $\mu(E_2) \leq \mu(E_1)$. Hence

$$\mu(CE_1) = \mu(CE_2).$$

□

(c) Uniqueness.

Let m_1, m_2 be extensions of μ as per the above extension. Now, $EUF \in \overline{\mathcal{M}}$. Observe

$$m_1(EUF) = \mu(CE) = m_2(EUF')$$

where F' is any subset of a null set.

Definition 0.13. An outer measure on a non-empty set X is a function $\mu^* \colon \mathcal{P}(X) \to [0, \infty]$ that satisfies,

1. $\mu^*(\emptyset) = 0$.
2. $\mu^*(A) \leq \mu^*(B)$ if $A \subseteq B$.
3. $\mu^* \left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Proposition 0.4. Let $\mathcal{E} \subseteq \mathcal{P}(X)$ and $\rho \colon \mathcal{E} \to \mathcal{P}(X)$ and $\rho \colon \mathcal{E} \to [0, \infty]$ be such that $\emptyset \in \mathcal{E}$, $X \in \mathcal{E}$ and $\rho(\emptyset) = 0$. For any $A \subset X$ we define

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid E_i \in \mathcal{E} \text{ and } A \subseteq \bigcup_{i=1}^{\infty} E_i \right\}.$$

Then μ^* is an outer measure.

Proof. (a) Observe a countable collection of empty sets covers $A = \emptyset$ hence

$$\mu^* \emptyset = \inf_{\substack{\mathcal{E} \\ \bigcup \mathcal{E} \supseteq \emptyset}} \sum_{E_n \in \mathcal{E}} \rho(E_n) \leq \sum_{n \in \mathbb{N}} \rho(\emptyset) = 0.$$

Also $\rho(E_n) \geq 0$ for any $E_n \in \mathcal{E}$ hence

$$\sum_{E_n \in \mathcal{E}} \rho(E_n) \geq 0 \quad \text{for any collection of intervals}$$

therefore

$$\inf_{\substack{\mathcal{E} \\ \bigcup \mathcal{E} \supseteq \emptyset}} \sum_{E_n \in \mathcal{E}} \rho(E_n) \geq 0.$$

So we get $\mu^* \emptyset = 0$.

For any non-empty set $A \subseteq X$ there exists $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{E}$ such that $A \subseteq \bigcup_{j=1}^{\infty} E_j$ (we can set $E_j = X$ for all j). □

Proof. (cb) If $A \subseteq B$ then if $\{E_j\}$ is a collection of sets covering B then $\mathcal{B}$ covers A as well. If $\mathcal{B}$ is a cover of B, we construct

$$\mathcal{A} = \{E_n \in \mathcal{B} \mid B_n \cap A \neq \emptyset\}$$

and observe $\mathcal{A} \subseteq \mathcal{B}$. Hence

$$\sum_{A_n \in \mathcal{A}} P(A_n) \leq \sum_{B_n \in \mathcal{B}} P(B_n).$$

Therefore

$$\mu^* A \leq \sum_{B_n \in \mathcal{B}} P(B_n).$$

Observe $\mu^* A \leq \mu^* B$; otherwise $\mu^* A > \mu^* B$ then there exist a countable collection $\{E_j\} = E$ covering B and also A whose sum, i.e. $\sum P(B_n)$, is less than $\mu^* A$, which contradicts the fact that $\mu^* A$ is the infimum of sums of countable collections of sets in E covering A. $\square$

Proof. (c) Suppose $\{A_j\}_{j=1}^{\infty} \subset \mathcal{P}(X)$ and fix $\delta > 0$. For each $j \in \mathbb{N}$ we get $\{E_{j,k}\}_{k=1}^{\infty} \subset \mathcal{E}$ such that

$$A_j \subset \bigcup_{k=1}^{\infty} E_{j,k} \quad \text{and} \quad \sum_{k=1}^{\infty} \rho(E_{j,k}) \leq \mu^*(A_j) + \frac{\delta}{2^j}.$$

But if $A = \bigcup_{j=1}^{\infty} A_j$, we have $A \subset \bigcup_{j,k=1}^{\infty} E_{j,k}$ and

$$\sum_{j,k} \rho(E_{j,k}) \leq \sum_{j} \rho(E_{j,k}) + \delta,$$

whence

$$\mu^*(A) \leq \sum_{j} \mu^*(A_j) + \delta.$$

As δ is arbitrary, we are done. $\square$ $\square$

Definition 0.14. * A set A is <u>measurable</u> if for each set E we have

$$m^* E = m^*(A \cap E) + m^*(A^c \cap E).$$

Note: We know for any set $m^* E \leq m^*(A \cap E) + m^*(A^c \cap E)$ holds. Hence A is measurable iff

$$m^* E \geq m^*(A \cap E) + m^*(A^c \cap E) \quad \text{for any set } E.$$

Remark 0.4. If A is measurable then A^c is measurable.

Lemma 0.3. If A_1 and A_2 are measurable sets then $A_1 \cup A_2$ is measurable.

Proof. For any set $E \subseteq X$,

$$\mu^*(E \cap (A_1 \cup A_2)) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2 \cap A_1^c)$$

and

$$\mu^*(E\cap(A_1\cup A_2))+\mu^*(E\cap(A_1\cup A_2)^c)=\mu^*(E\cap A_1)+\mu^*(E\cap A_2\cap A_1^c)+\mu^*(E\cap A_2^c\cap A_1^c).$$

Now apply the fact that A_2 is a measurable set,

$$\mu^*(E\cap(A_1\cup A_2))+\mu^*(E\cap(A_1\cup A_2)^c)=\mu^*(E\cap A_1)+\mu^*(E\cap A_1^c)=\mu^*(E)$$

as A_1 is a measurable set. Hence $A_1\cup A_2$ is measurable. □

Lemma 0.4. Let E be any set, and $A_1,\ldots,A_n$ a finite sequence of disjoint measurable sets. Then

$$\mu^*\left(E\cap\left(\bigcup_{i=1}^n A_i\right)\right)=\sum_{i=1}^n\mu^*(E\cap A_i).$$

Proof. Base case holds for $n=1$. Assume the given holds for $n-1$ sets.

Observe

$$E\cap\left(\bigcup_{i=1}^{n-1}A_i\right)\cap A_n=E\cap A_n$$

and

$$E\cap\left(\bigcup_{i=1}^{n-1}A_i\right)\cap A_n^c=E\cap\bigcup_{i=1}^{n-1}A_i$$

as A_n is disjoint from $\{A_i\}_{i=1}^{n-1}$. Hence

$$\mu^*\left(E\cap\left(\bigcup_{i=1}^n A_i\right)\right)=\mu^*\left(E\cap\bigcup_{i=1}^{n-1}A_i\cap A_n\right)+\mu^*\left(E\cap\bigcup_{i=1}^{n-1}A_i\cap A_n^c\right)$$

by measurability of A_n. This implies

$$\mu^*\left(E\cap\left(\bigcup_{i=1}^n A_i\right)\right)=\mu^*(E\cap A_n)+\mu^*\left(E\cap\bigcup_{i=1}^{n-1}A_i\right).$$

□

By induction hypothesis, we get

$$\mu^*\left(E\cap\left(\bigcup_{i=1}^n A_i\right)\right)=\sum_{i=1}^n\mu^*(E\cap A_i).$$

Theorem 0.2 (Carathéodory's Theorem)**.** If μ^* is an outer measure on X, the collection $\mathcal{M}$ of μ^*-measurable sets is a σ-algebra and the restriction of μ^* to $\mathcal{M}$ is a complete measure.

Proof. According to a previous remark and lemma, $\mathcal{M}$ is an algebra and finitely additive. As per the previous lemma,

$$\mu^*\left(E \cap \left(\bigcup_{i=1}^{\infty} A_i\right)\right) = \sum_{i=1}^{\infty} \mu^*(E \cap A_i)$$

holds for any $E \subset X$ and collection of μ^*-measurable disjoint sets $\{A_i\}_{i=1}^{\infty}$. $\square$

To ensure the content is properly structured and formatted according to the rules, —

In order to show that $\mathcal{M}$ is closed under disjoint unions, let us proceed as follows:

Let $B_n = \bigcup_{j=1}^{n} A_j$ and $B = \bigcup_{j=1}^{\infty} A_j$. Hence,

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \\ &\geq \sum_{j=1}^{n} \mu^*(E \cap A_j) + \mu^*(E \cap B^c). \end{aligned}$$

Observe the statement above holds for any arbitrary n. Therefore,

$$\begin{aligned} \mu^*(E) &\geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(E \cap \left(\bigcup_{j=1}^{\infty} A_j\right)\right) + \mu^*(E \cap B^c) \\ &= \mu^*(E \cap B) + \mu^*(E \cap B^c) \geq \mu^*(E). \end{aligned}$$

It follows that $B \in \mathcal{M}$ and $\mu^*(B) = \sum_{j=1}^{\infty} \mu^*(A_j)$, so μ^* is countably additive, by substituting $E = B$.

Finally, if $\mu^*(A) < 0$ for any $E \subset X$, we have

$$\begin{aligned} \mu^*(E) &\leq \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A^c) \leq \mu(E). \end{aligned}$$

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So, $A \in \mathcal{M}$. Hence $\mu^*|_{\mathcal{M}}$ is complete measure.

Definition 0.15. Let $A \subset \mathcal{P}(X)$ be an algebra, a function $\mu_0 \colon A \to [0, \infty]$ will be called a premeasure if

1. $\mu_0(\emptyset) = 0$,
2. if $\{A_j\}_{j \in \mathbb{N}}$ is a sequence of disjoint sets in A, then

$$\mu_0\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} \mu_0(A_j).$$

Remark 0.5. Note: The difference between a measure and a premeasure is the domain.

- Measure is defined on a σ-algebra.
- Premeasure is defined on an algebra.

Remark 0.6. We have a notion of finite and σ-finite premeasures as same as that of a measure.

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Remark 0.7. If μ_0 is a premeasure on $\mathcal{A} \subseteq \mathcal{P}(X)$, it induces an outer measure on X in accordance with Proposition **??**, namely

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(A_j) \mid A_j \in \mathcal{A},\, E \subseteq \bigcup_{j=1}^{\infty} A_j \right\}.$$

Lemma 0.5. If μ_0 is a premeasure on $\mathcal{A}$ and μ^* as defined in the remark above, then

1. $\mu^*|_{\mathcal{A}} = \mu_0$;
2. every set in $\mathcal{A}$ is μ^*-measurable.

Proof. (a) Suppose $E \in \mathcal{A}$. If $E \subseteq \bigcup_{j=1}^{\infty} A_j$ with any countable collection $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{A}$, then we let $B_n = E \cap \left(A_n \setminus \bigcup_{j=1}^{n-1} A_j \right)$. Observe B_n is a disjoint member of $\mathcal{A}$ whose union is E, so □

—

$$\mu_0(E) = \sum_{n=1}^{\infty} \mu_0(B_n) \leq \sum_{n=1}^{\infty} \mu_0(A_j).$$

Since this holds for any arbitrary cover of E, hence as per the definition of μ^*, we get

$$\mu_0(E) \leq \mu^*(E).$$

Now as $E \in \mathcal{A}$, we can choose a cover $E \subset \bigcup_j A_j$ s.t. $A_1 = E$ and $A_j = \emptyset$ for $j > 1$. Hence $\mu_0(E) = \mu^*(E)$.

Proof. 1. If $A \in \mathcal{T}$, $E \subset X$ and $\varepsilon > 0$, then there is a sequence $\{B_j\}_{j \in \mathbb{N}} \subset \mathcal{A}$ with

$$E \subset \bigcup_{j=1}^{\infty} B_j \quad \text{and} \quad \sum_{j=1}^{\infty} \mu_0(B_j) \leq \mu^*(E) + \varepsilon.$$

Since μ_0 is additive on $\mathcal{A}$,

$$\mu_0 \left(\bigcup_{j=1}^{\infty} B_j \cap X \right) = \mu_0 \left(\bigcup_{j=1}^{\infty} B_j \cap (A \cup A^c) \right)$$

$$= \mu_0 \left(\bigcup_{j=1}^{\infty} (B_j \cap A) \right) + \mu_0 \left(\bigcup_{j=1}^{\infty} (B_j \cap A^c) \right)$$

$$= \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c).$$

Hence,

$$\begin{aligned} \mu^*(E) + \varepsilon &\geq \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c) \\ &\geq \mu_0(E \cap A) + \mu_0(E \cap A^c). \end{aligned}$$

□

—

As the set E is arbitrary, A is μ^*-measurable.

Theorem 0.3. Let $A \subseteq \mathcal{P}(X)$ be an algebra, μ_0 is a premeasure defined on A. Let $\mathcal{M}$ be σ-algebra generated by A. There exists a measure μ on $\mathcal{M}$ such that $\mu|_A = \mu_0$.

Proof. If we define μ^* as defined earlier and $\mu := \mu^*|_{\mathcal{M}}$. Then we know by the previous lemma $\mu^*|_A = \mu|_A = \mu_0$ and all A are μ^*-measurable. This implies σ-algebra of μ^*-measurable sets include A. □

Theorem 0.4. If ν is another measure defined on $\mathfrak{M}$ (as per the previous theorem) that extends on μ_0, then $\nu(E) \leq \mu(E)$ for any $E \in \mathfrak{M}$ and equality when $\mu(E) < +\infty$. If μ_0 is σ-finite, then μ is the unique extension of μ_0 to a measure on $\mathfrak{M}$.

Proof. If $E \in \mathfrak{M}$ and $E \subset \bigcup_{j=1}^{\infty} A_j$ where $A_j \in \mathcal{A}$ for all $j \in \mathbb{N}$, then

$$\nu(E) \leq \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$$

holds true for any $\{A_j\}_{j=1}^{\infty}$ collection. Hence,

$$\nu(E) \leq \mu(E).$$

Let $A = \bigcup_{j=1}^{\infty} A_j$. Now

$$\nu(A) = \nu\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} \nu(A_j) = \lim_{n\to\infty} \sum_{j=1}^{n} \nu(A_j) = \lim_{n\to\infty} \nu\left(\bigcup_{j=1}^{n} A_j\right).$$

Since ν extends μ_0,

$$= \lim_{n\to\infty} \mu_0\left(\bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{\infty} \mu_0(A_j) = \mu_0(A).$$

□

—

If $\mu(E) < +\infty$. For every $\varepsilon > 0$ there exist a collection of sets $\{A_\varepsilon, j\}_{j=1}^{\infty}$, such that $A_\varepsilon = \bigcup_{j=1}^{\infty} A_{\varepsilon j}$ and $\mu(A_\varepsilon) < \mu^*(E) + \varepsilon$, hence $\mu(A_\varepsilon \setminus E) < \varepsilon$, and $\mu^*(CE) \leq \mu(A) = \nu(A) = \nu(E) + \nu(A \setminus E) \leq \nu(CE) + \mu(A \setminus E) \leq \nu(CE) + \varepsilon$ holds for all $\varepsilon > 0$. This implies $\mu^*(CE) \leq \nu(CE)$.

Proof. Finally suppose $X = \bigcup_{j=1}^{\infty} A_j$ with $\mu_0(A_j) < \infty$ where we can assume A_j's are disjoint (or reconstruct a given collection into that way). Then for any $E \in \mathcal{M}$

$$\mu(E) = \sum_{j=1}^{\infty} \mu(E \cap A_j) = \sum_{j=1}^{\infty} \nu^*(E \cap A_j) = \nu^*(CE),$$

so $\mu = \nu$. □

—

Definition 0.16. Let $(a, b]$ be denoted as h-intervals.

Let μ be the Borel measure on $\mathcal{B}_{\mathbb{R}}$. We define the distribution function as

$$F(x) = \mu((-\infty, x]).$$

Proposition 0.5. Let $F : \mathbb{R} \to \mathbb{R}$ be an increasing and right continuous function. If $\{(a_j, b_j]\}_{j=1}^{n}$ are disjoint h-intervals, let

$$\mu_0 \left(\bigcup_{j=1}^{n} (a_j, b_j] \right) = \sum_{j=1}^{n} \mu_0((a_j, b_j]),$$

and let $\mu_0(\emptyset) = 0$. Then μ_0 is a premeasure on algebra $\mathcal{A}$.

Proof. (a) μ_0 is well defined.

Given any collection of disjoint h-intervals, if $\{(a_j, b_j]\}_{j=1}^{n}$ by relabelling, we get

$$a = a_1 < b_1 = a_2 < \cdots < b_n = b.$$

This implies

$$\sum_{j=1}^{n} [F(b_j) - F(a_j)] = F(b) - F(a).$$

More generally, if $\{I_i\}_{i=1}^{n}$ and $\{J_j\}_{j=1}^{m}$ are finite sequences of disjoint h-intervals such that $\bigcup_{i=1}^{n} I_i = \bigcup_{j=1}^{m} J_j$, by the reasoning above we can say,

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j).$$

Proof. (b) μ_0 is countably additive

Observe μ_0 is finitely additive by construction. We want to show if $\sum I_i \in \mathcal{U}$ s.t. $\bigcup_{i\in\mathbb{N}} I_i \in \mathcal{U}$ then

$$\mu_0\left(\bigcup_{i=1}^{\infty} I_i\right) = \sum_{i=1}^{\infty} \mu_0(I_i).$$

As $\bigcup_{i=1}^{\infty} I_i \in \mathcal{U}$, we can separate $\{I_i\}_{i=1}^{\infty}$ into finitely many subsequences whose union is an h-interval. Assume $\bigcup_{i=1}^{n} I_i$ is an h-interval (otherwise you look into a smaller collection of I_i's whose union is an h-interval.)

Let $I = \bigcup_{i=1}^{\infty} I_i = [a, b)$. Observe

$$\mu_0(I) = \mu_0\left(\bigcup_{j=1}^{n} I_j\right) + \mu_0\left(I \setminus \bigcup_{j=1}^{n} I_j\right)$$

$$\geq \mu_0\left(\bigcup_{j=1}^{n} I_j\right) = \sum_{j=1}^{n} \mu_0(I_j).$$

□

To show the reverse inequality, consider the following case:

Proof. **Case (ci):** If a, b are finite.

As F is right continuous, for any $\varepsilon > 0$, there exists $\delta_\varepsilon > 0$ such that

$$F(a + \delta_\varepsilon) - F(a) < \varepsilon.$$

If $I_j = (a_j, b_j]$ for $j \in \mathbb{N}$, then we can find a $\delta_{\varepsilon,j} > 0$ such that

$$F(b_j + \delta_{\varepsilon,j}) - F(b_j) < \frac{\varepsilon}{2^j}.$$

Now, $\{(a_j, b_j + \delta_{\varepsilon,j})\}_{j=1}^{\infty}$ is a collection of open intervals covering $[a + \delta_\varepsilon, b]$, so there is a finite subcover. Let $(a_1, b_1 + \delta_1), \ldots, (a_N, b_N + \delta_N)$ cover $[a + \delta_\varepsilon, b]$ and $b_i + \delta_i \in (a_{i+1}, b_{i+1} + \delta_{i+1})$. (We have replaced $\delta_{\varepsilon,i} = \delta_i$ for all $i = \{1, \ldots, N-1\}$.)

For any $\varepsilon > 0$, observe

$$\mu_0(I) < F(b) - F(a + \delta_\varepsilon) + \varepsilon$$

$$\leq F(b_N + \delta_N) - F(a_1) + \varepsilon.$$

□

$$= F(b_N + \delta_N) - F(a_N) + \sum_{i=1}^{N-1} [F(a_{j+1}) - F(a_j)] + \varepsilon$$

$$\leq F(b_N + \delta_N) - F(a_N) + \sum_{i=1}^{N} [F(b_j + \delta_j) - F(a_j)] + \varepsilon$$

$$< \left(\sum_{i=1}^{N} F(b_j) + \frac{\varepsilon}{2^j} - F(a_j) \right) + \varepsilon < \sum_{i=1}^{\infty} \mu(I_j) + 2\varepsilon.$$

As the above statement holds for all $\varepsilon > 0$, we are done.

(ii) If one of them is NOT finite.

Say $a = -\infty$. Fix $\varepsilon > 0$. Then for any $M > 0$, we can find an open cover $(a_j + b_j + S_j)$ covering $[-M, b]$. Hence by the same reasoning we get

$$F(b) - F(-M) \leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon,$$

whereas if $b = \infty$, for any $M < \infty$ we get

$$F(M) - F(a) \leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon.$$

We get the desired result by $\varepsilon \to 0$ followed by $M \to \infty$. (Go through this).

Theorem 0.5. If $F : \mathbb{R} \to \mathbb{R}$ is any increasing, right continuous function, tpremeasure$_induction.ItisclearthatF and G$
- G$isconstant, andthesepremeasuresare\sigma$-finite as $\mathbb{R} = \bigcup_{j\in\mathbb{Z}}(j, j+1]$. □

The first two assertions therefore follow from **??**.

As for the converse, the monotonicity of μ implies F to be an increasing function.

Observe μ is continuous from above and below implies the right continuity of F for $x \geq 0$ and $x < 0$.

It is evident that $\mu = \mu_F$ on $\mathcal{U}$, and hence $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$ by uniqueness of **??**.

Remark 0.8. 1. The theory could have also been developed for $[a, b)$ intervals and F as left continuous.

2. If μ is a finite Borel measure on $\mathbb{R}$ then

$$F(x) = \mu((-\infty, x])$$

is the cumulative distribution function.

3. Any increasing right continuous F gives us a complete measure.

We denote this complete measure whose domain includes $\mathcal{B}_{\mathbb{R}}$ as Lebesgue–Stieltjes measure associated to F.

Given an increasing, right continuous function F, we get $(\mathbb{R}, \mathcal{M}_\mu, \mu)$ as a complete measure space where for any $E \in \mathcal{M}_\mu$,

$$\mu(E) = \inf\left\{\sum_{j=1}^{\infty}[F(b_j) - F(a_j)] \mid E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j]\right\}.$$

Lemma 0.6. For any $E \in \mathcal{M}_\mu$,

$$\mu(E) = \inf\left\{\sum_{j=1}^{\infty}\mu((a_j, b_j]) \mid E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j]\right\}.$$

Proof. Let $\nu\colon \mathcal{M}_\mu \to [0, \infty)$ such that

$$\nu(E) = \inf\left\{\sum_{j=1}^{\infty}\mu((a_j, b_j]) \mid E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j]\right\}$$

and μ be the measure defined in the above given definition.

Goal: $\nu(E) = \mu(E)$ for all $E \in \mathcal{M}_\mu$. □

—

1. We want to show $\mu(E) \leq \nu(E)$ for any $E \in \mathcal{M}_\mu$.

   Suppose $E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j)$ and for each (a_j, b_j) we get $\{I_j^k\}_{k=1}^{\infty}$ as disjoint h-intervals covering (a_j, b_j) for each k. Let $I_j^k = [c_j^k, c_j^{k+1}]$ where $c_j^1 = a_j$ and $c_j^k \to b_j$ as $k \to \infty$. Thus $E \subseteq \bigcup_{j,k} I_{j,k}$. Hence
   $$\sum_{j=1}^{\infty}\mu((a_j, b_j)) = \sum_{j=1}^{\infty}\mu(I_j^k) \geq \mu(E).$$
   (Or you can use $E \subseteq \bigcup_{j\in\mathbb{N}}(a_j, b_j)$ and by monotonicity and subadditivity and apply infimum, we get $\mu(E) \leq \nu(E)$.)

   Proof. Since $E \subseteq \bigcup_{j,k} I_{j,k}$ and $\{I_{j,k}\}$ are disjoint, by the subadditivity of μ we have
   $$\mu(E) \leq \sum_{j,k}\mu(I_{j,k}).$$
   For each j, since $\{I_j^k\}_{k=1}^{\infty}$ covers (a_j, b_j) and μ is monotone, we have
   $$\mu((a_j, b_j)) \leq \sum_{k=1}^{\infty}\mu(I_j^k).$$

Summing over j gives

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) \leq \sum_{j,k} \mu(I_{j,k}).$$

Since $\mu(E) \leq \sum_{j,k} \mu(I_{j,k})$ and $\sum_{j=1}^{\infty} \mu((a_j, b_j))$ is the infimum over all such coverings, it follows that $\mu(E) \leq \nu(E)$. $\square$

2. We want to show $\nu(E) \leq \mu(E)$; for any $\varepsilon > 0$, we get $\{(a_j, b_j)\}_{j=1}^{\infty}$ covering E and

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) \leq \mu(E) + \varepsilon.$$

—

As F is right continuous and increasing, for each $\varepsilon > 0$, we find $\delta_{\varepsilon j} = \delta_j$ for each $j \in \mathbb{N}$ such that

$$F(b_j + \delta_j) - F(b_j) < \frac{\varepsilon}{2^j}.$$

Then $E \subset \bigcup_{j=1}^{\infty} (a_j + b_j + \delta_j)$ and

$$\begin{aligned} \mu((a_j, b_j + \delta_j)) \leq \mu((a_j, b_j + \delta_j)) &= F(b_j + \delta_j) - F(a_j) \\ &= F(b_j + \delta_j) - F(b_j) + F(b_j) - F(a_j) \end{aligned}$$

for all $j \in \mathbb{N}$. Now by taking a sum,

$$\begin{aligned} \sum_{j=1}^{\infty} \mu((a_j, b_j)) &\leq \sum_{j=1}^{\infty} [F(b_j + \delta_j) - F(b_j)] + \sum_{j=1}^{\infty} \mu((a_j, b_j)) \\ &\leq \varepsilon + \varepsilon + \mu(E). \end{aligned}$$

Proof. Observe that $\nu(E) \leq \sum_{j=1}^{\infty} \mu((a_j, b_j)) \leq \mu(E) + 2\varepsilon$ holds for all $\varepsilon > 0$. Hence $\nu(E) \leq \mu(E)$. $\square$

Theorem 0.6. If $E \in \mathcal{M}_\mu$, then

$$\begin{aligned} \mu(E) &= \inf \{\mu(U) \mid U \supseteq E \text{ and } U \text{ is open}\} \\ &= \sup \{\mu(K) \mid K \subseteq E \text{ and } K \text{ is compact}\}. \end{aligned}$$